Department of Physics

FSK116 - **MEMO**
June Examination

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Natural and Agricultural Sciences
Fakulteit Natuur- en Landbouwetenskappe
Lefapha la Disaense tša Tlhago le Temo

Examiner: Mr MJ Legodi
Moderator: Mr RQ Odendaal

Please read and understand the following.

1. All test and examination regulations of the University of Pretoria apply.
2. This is an OPEN BOOK assignment. You may use any source available to you, like your textbook and lecture notes, to assist in answering the questions. However, you **MAY NOT** receive (or provide) help in any way whatsoever from (or to) any other person. This includes the lecturer, tutor, a friend, family members, paid-for or free assignment assistance services, etc.
3. Answer the questions **in your own handwriting** on separate paper, lined or unlined.
   Take quality photographs of your answer sheets and scan them as a PDF document. Use apps like **CamScanner** to convert the photographs to PDF.
   Submit your answer sheets, via ClickUP, in the "**Exams Folder**" under the "**FSK116 Part B June Exam (Take-Home) Submission**". **You need to submit before 15:15.**
   There will be a queue during submission. You may not get your submission receipt immediately! To accommodate this, we give you a grace period of 30 minutes between 15:15 and 15:45 only, to make your submission without attracting any penalties. However, your submission will still be tagged "late".
   YOU HAVE BEEN WARNED: Late submissions may not be accepted and will be penalised!
4. Number and label your submissions as in the question paper. Incorrectly-numbered and incorrectly-labelled answers may not be marked.
5. Provide neatly labelled sketches, including co-ordinate systems, free-body diagrams, final answers with correct numbers of significant figures, units, etc. Failure to do so will result in marks being deducted.
6. Your answer needs to be written in a systematic and logical fashion. Map out the steps you will take to reach your goal i.e. solve the problem in your mind first! You must show that you understand and can use the relevant physics. Only substitute numerical values in the final step of calculations! A correct answer does not necessarily mean that you will receive full marks!
7. Units must be included when numbers are substituted; else, marks will be deducted.
8. You will need to look up constants and typical values for parameters in your calculations.
9. If any misconduct with respect to this examination is suspected, you will receive a mark of zero and will have to attend a formal disciplinary hearing, which may result in your expulsion from the University of Pretoria.

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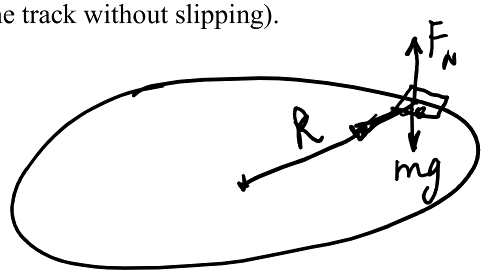
UNIVERSITY OF PRETORIA - Department of Physics

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of 2 the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I understand what plagiarism is and am aware of the University's policy in this regard. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

This statement is a reminder of the Code of Conduct that you have already signed. If you continue with this examination, you accept all the relevant rules.

QUESTION 1 [15]

1.1 The coefficients of friction between a race car's tyres and the track surface are 0.950 for static friction and 0.800 for kinetic friction. The car is on a level circular track of radius 50.0 m on a planet where $g = 2.45\ m/s^2$ compared to Earth's $g = 9.80\ m/s^2$. Compare the maximum safe speed for the race car on the track on the planet and the maximum safe speed on Earth. (ie. compare the maximum speeds that the race car can go around the track without slipping). (3)



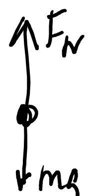
$$\sum F_r = \frac{mv^2}{R}$$ NII radial

$$f_s = \frac{mv^2}{R}$$

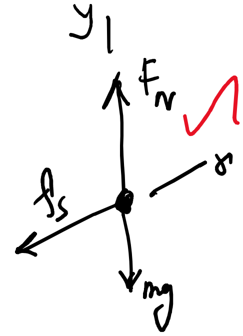
$$f_s = \mu_s (mg) = \frac{mv^2}{R}$$

$$\mu_s g = \frac{v^2}{R}$$

$$v_{max} = \sqrt{\mu_s g R}$$



$\sum F_y = 0$ NII



Earth

$$v_{max,E} = \sqrt{0.950(9.80\ m/s^2)(50.0\ m)} = 21.575\ m/s$$

Planet

$$v_{max,Pl} = \sqrt{0.950(2.45\ m/s^2)(50.0\ m)} = 10.7877\ m/s$$

$$v_{max,E} = 2 v_{max,Planet}$$

3

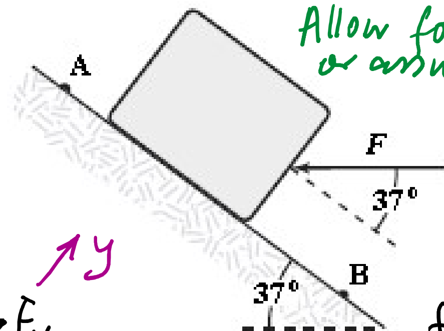
1.2 A block of mass *m* is lowered down a surface inclined at *θ* degrees for a distance *d* between points A and B. A horizontal force F is acting on the block between A and B. Take the kinetic energy of the block at points A and B as K_A and K_B, respectively.

(a) Construct an analytic expression to determine how much work is done on the block by the force of friction between points A and B. (6)

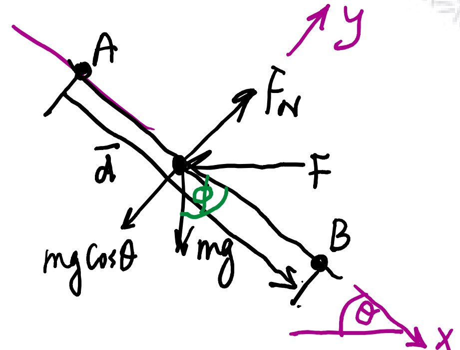
(b) Now determine the actual amount of work done in (a) above, if the block has a mass of 4.0 kg, the angle of inclination, is $\theta = 37°$, the distance $d = 5.0$ m, $K_A = 10$ J and $K_B = 20$ J. $F = 10\,N$ (2)

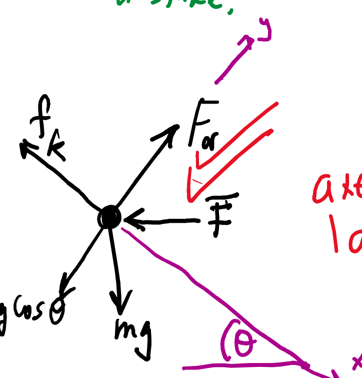
The value for F was announced when the exam was in session. Allow for own (student supplied) or assumed F values.

Also, for unknown values for F. Only 1 mark is at stake!



(a)

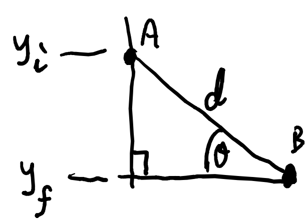




axes, labels

$$W_{others} = \Delta K + \Delta U + \Delta E_{int} \checkmark \quad [\text{Earth, block, surface} = \text{system}]$$

$$\vec F\cdot\vec d + \vec F_N\cdot\vec d \checkmark = (K_f - K_i) + (mgy_f - mgy_i) + f_k d \quad [\Delta E_{int} = f_k d] \text{ — (A)}$$

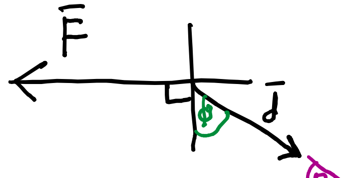


$\vec F$ and F_N are constant

$\vec F_N\cdot\vec d = 0$ since the angle between is zero.

$|y_f - y_i| = d\sin\theta$ and define $y_f = 0$

$|y_i| = d\sin\theta$ ✓



$$\vec F\cdot\vec d = Fd\cos(90°+\phi)$$
$$= -Fd\sin\phi$$
$$= -Fd\cos\theta \checkmark$$

+1 BONUS POINT IF DETAILS ARE SUPPLIED.

$\boxed{6}$

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From (A) above:

$$-Fd\cos\theta + 0 = (K_B - K_A) + (0 - mgy_i) + f_k d$$

$$\therefore f_k d = mgy_i + K_A - K_B - Fd\cos\theta \checkmark$$

(b) $= (4.0\,kg)(9.80\,m/s^2)(5.0\,m\sin 37°) + 10J - 20J - (10\,N)(5.0\,m)\cos 37°$ ✓ units

$= 68\,J$ ✓ correct answer

$\boxed{2}$

1.3 If the resultant force acting on a 2.0-kg object is equal to $\left(3\hat{\mathbf{i}}+4\hat{\mathbf{j}}\right)$ N, what is the change in kinetic energy as the object moves from $\left(7\hat{\mathbf{i}}-8\hat{\mathbf{j}}\right)$ m to $\left(11\hat{\mathbf{i}}-5\hat{\mathbf{j}}\right)$ m? (4)

$W_{net} = \Delta K$ Work Kinetic Energy Theorem.

$$\Delta K = W_{net}$$
$$= (3\hat{i}+4\hat{j})\cdot[(11\hat{i}-5\hat{j})-(7\hat{i}-8\hat{j})] \text{ N.m}$$
$$= (3\hat{i}+4\hat{j})\cdot(4\hat{i}+3\hat{j}) \text{ N.m}$$
$$= 12+12 \text{ N.m}$$
$$= 24 \text{ N.m}$$

4

QUESTION 2 [13]

2.1 A wheel rotating about a fixed axis with a constant angular acceleration of 2.0 rad/s^2 turns through 2.4 revolutions during a 2.0 s time interval. What is the angular velocity at the end of this time interval? (4)

$$\Delta\theta = \omega_i t + \frac{1}{2}\alpha t^2 \checkmark$$

$$\omega_i = \frac{\Delta\theta - \frac{1}{2}\alpha t^2}{t}$$

$$= \frac{2.4\,\text{rev} \times \frac{2\pi\,\text{rad}}{\text{rev}} \checkmark - \frac{1}{2}(2.0\,\text{rad/s}^2)(2.0\,\text{s})^2}{2.0\,\text{s}}$$

$$= 5.539822\,\text{rad/s} \checkmark$$

$$\Rightarrow \omega_f = \omega_i + \alpha t$$

$$= 5.539822\,\text{rad/s} + (2.0\,\text{rad/s}^2)(2.0\,\text{s})$$

$$= \underline{\underline{9.5\,\text{rad/s}}} \checkmark$$

2.2 The linear mass density of a rod, in g/m, is given by the expression $\lambda = 40.0 + 30.0x$, where x is a distance measured in meters. The rod extends from the origin to $x = 0.400$ m. What is the location of the center of mass of the rod? (4)

$$x_{cm} = \frac{\int x\,dm}{\int dm}$$

$$= \frac{\int x \lambda dx}{\int \lambda dx}$$ ✓ ✓

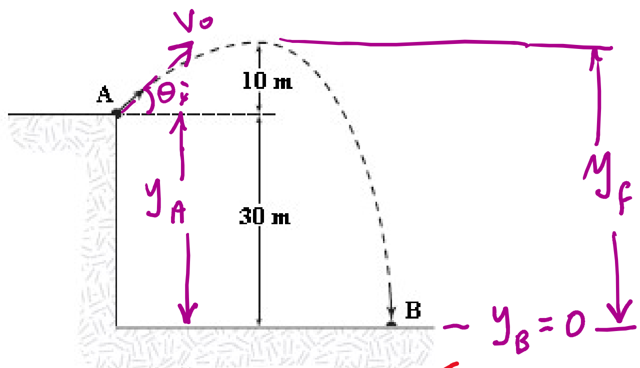
$$= \frac{\int_0^{0.400} x(40.0 + 30.0x)dx}{\int_0^{0.400} (40.0 + 30.0x)\,dx}$$ ✓ ✓

$$= \frac{\left[20.0x^2 + 10.0x^3\right]_0^{0.400}}{\left[40.0x + 15.0x^2\right]_0^{0.400}}$$

4

$$= 0.209 \text{ m}$$ ✓

2.3 A 0.040 kg ball is thrown from the top of a 30 m tall building (point A) at an unknown angle above the horizontal (You may assume the unknown angle to be θ). As shown in the figure, the ball attains a maximum height of 10 m above the top of the building before striking the ground at point B. If air resistance is negligible, what is the value of the kinetic energy of the ball at point B minus the kinetic energy of the ball at point A ie. $(K_B - K_A)$? (5)



$E = K + U$ for the isolated ball-Earth system with origin at $y_B = 0$.

$E_{top} = \frac{1}{2}m(V_0 \cos\theta)^2 + mgy_f$ and it is constant.

$E_{total} = \frac{1}{2}mv_B^2 + (mgy_i = 0)$

$= \frac{1}{2}mV_{B,y}^2 + \frac{1}{2}m(V_0 \cos\theta_i)^2$ just before hitting ground. And is constant!

It is not necessary to explicitly give the expressions for K_A & K_B.

$E_{at\,A} = \frac{1}{2}mV_0^2 + mgy_A$ and is constant.

All the E's are the same.

$\therefore K_B - K_A = E_{Total,B} - E_{Total,A} + mgy_A$

$= mgy_A = (0.040\text{ kg})(9.80\text{ m/s}^2)(30\text{ m})$

$= 12\text{ J}$

5

Alternative:

At the top: $E = K + U = \frac{1}{2}m(V_0 \sin\theta)^2 + mgy_f$

At point B: $K_B = \frac{1}{2}mv_B^2 = E$

At point A: $K_A + U_A = E$

$\therefore K_A = \frac{1}{2}mv_0^2 = E - mgy_A$

$\Rightarrow K_B - K_A = E - E + mgy_A$

$= (0.040\text{ kg})(9.80\text{ m/s}^2)(30\text{ m})$

$= 12\text{ J}$

QUESTION 3 [14]

3.1 The motion of a particle connected to a spring is described by $x(t) = 10 \sin(\pi t)$. At what time (in s) is the potential energy of the particle equal to its kinetic energy? (3)

$E = K + U$ ✓

$\frac{1}{2}kA^2 = 2\left(\frac{1}{2}kA^2 \sin^2(\omega t)\right)$. ✓ We can safely assume ϕ in $(\omega t + \phi)$ to be equal to zero.

$1 = 2\sin^2(\omega t)$

$\Rightarrow \sin(\omega t) = \sqrt{\frac{1}{2}}$

$\Rightarrow \omega t = \sin^{-1}\left(\sqrt{\frac{1}{2}}\right) = \frac{\pi}{4}$

$\therefore \quad t = \left(\frac{\pi}{4}\right)\left(\frac{1}{\pi}\right) = 0.25\text{ s}$ ✓

3

Alternative:

$$\cancel{\frac{1}{2}kA^2} \cos^2(\omega t + \phi) = \cancel{\frac{1}{2}kA^2} \sin^2(\omega t + \phi)$$ ✓

Divide by $\cos^2(\omega t + \phi)$: $\frac{\cos^2(\omega t + \phi)}{\cos^2(\omega t + \phi)} = \frac{\sin^2(\omega t + \phi)}{\cos^2(\omega t + \phi)}$

$1 = \tan^2(\omega t + \phi)$ ✓

We can drop ϕ as before:

$\Rightarrow \tan^2(\omega t) = 1$

$\therefore (\omega t) = \tan^{-1}(\sqrt{1})$

$\Rightarrow t = \left(\frac{\pi}{4}\right)\left(\frac{1}{\pi}\right) = 0.25\text{ sec.}$ ✓

3

3.2 A 66.0 cm long string fixed at both ends is vibrating such that it forms a standing wave with three antinodes.

(a) Which harmonic does this wave represent? (3) Third Harmonic (1)

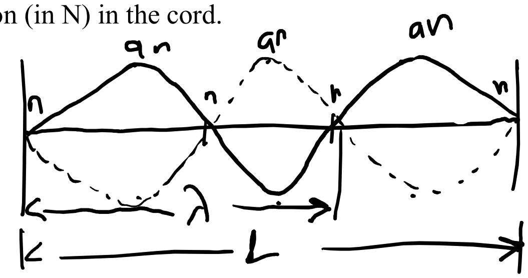
(b) Determine the wavelength (in cm) of this wave. $\lambda = 44.0$ cm (1)

(c) How many nodes are there in the wave pattern? 4 (1)

(d) If the string has a linear mass density of 0.00526 kg/m and is vibrating at a frequency of 261.6 Hz, determine the tension (in N) in the cord. (2)

General formula

(a) $L = n\frac{\lambda}{2}$, $n = 1, 2, 3, \ldots$



We can see that $n = 3$, implying a third harmonic.

Accept qualitative answers for (a) and (c)

From $L = n\frac{\lambda}{2}$ with $n=3$

(b) $\Rightarrow L = \frac{3}{2}\lambda$

$\Rightarrow \lambda = \left(\frac{2}{3}\right)(66.0\text{ cm})$

$= 44.0$ cm

(c) nodes = 4

5

(d) $v = \sqrt{\frac{T}{\mu}} \Rightarrow T = \mu v^2$

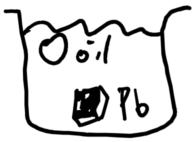
$= \mu(\lambda f)^2$

$= (5.26 \times 10^{-3}\text{ kg/m})[(44.0 \times 10^{-2}\text{ m})(261.6\text{ Hz})]^2$

$= 69.7$ N

3.3 You wish to cool a 1.95 kg block of lead initially at 88.0 °C to a temperature of 57.0 °C by placing it in a container of olive oil initially at 20.0 °C.

Determine the volume (in litres) of the liquid needed in order to accomplish this task. The density and specific heat capacity of olive oil are 860 kg/m^3 and 1970 J/(kg · °C), respectively (6)



$M_0 = ?$

$V_0 = ?$

	Lead	Olive oil
Mass	1.95 kg	?
Specific Heat Capacity, c	128 J/kg·°C	1970 J/kg·°C
T_i	88.0°C	20.0°C
T_f	57.0°C	57.0°C
ρ		860 kg/m^3

$$\Delta Q = 0 \Rightarrow Q_{cold} = -Q_{hot}$$

Pb and oil constitute an isolated system

$$m_0 c_0 \Delta T_0 = -m_{Pb} c_{Pb} \Delta T_{Pb}$$

$$m_0 = \frac{-m_{Pb} c_{Pb} \Delta T_{Pb}}{c_0 \Delta T_0}$$

$$= \frac{(-1.95 kg)(128 J/kg\cdot°C)(57.0°C - 88.0°C)}{(1970 J/kg\cdot°C)(57.0°C - 20.0°C)}$$

$$\simeq 0.10615 kg$$

6

From $\rho = \frac{mass}{Volume}$

$$\Rightarrow V_0 = \frac{m}{\rho} = \frac{0.10615 kg}{860 kg/m^3} \cdot \frac{liter}{10^{-3} m^3} = 0.123 \text{ liter}$$